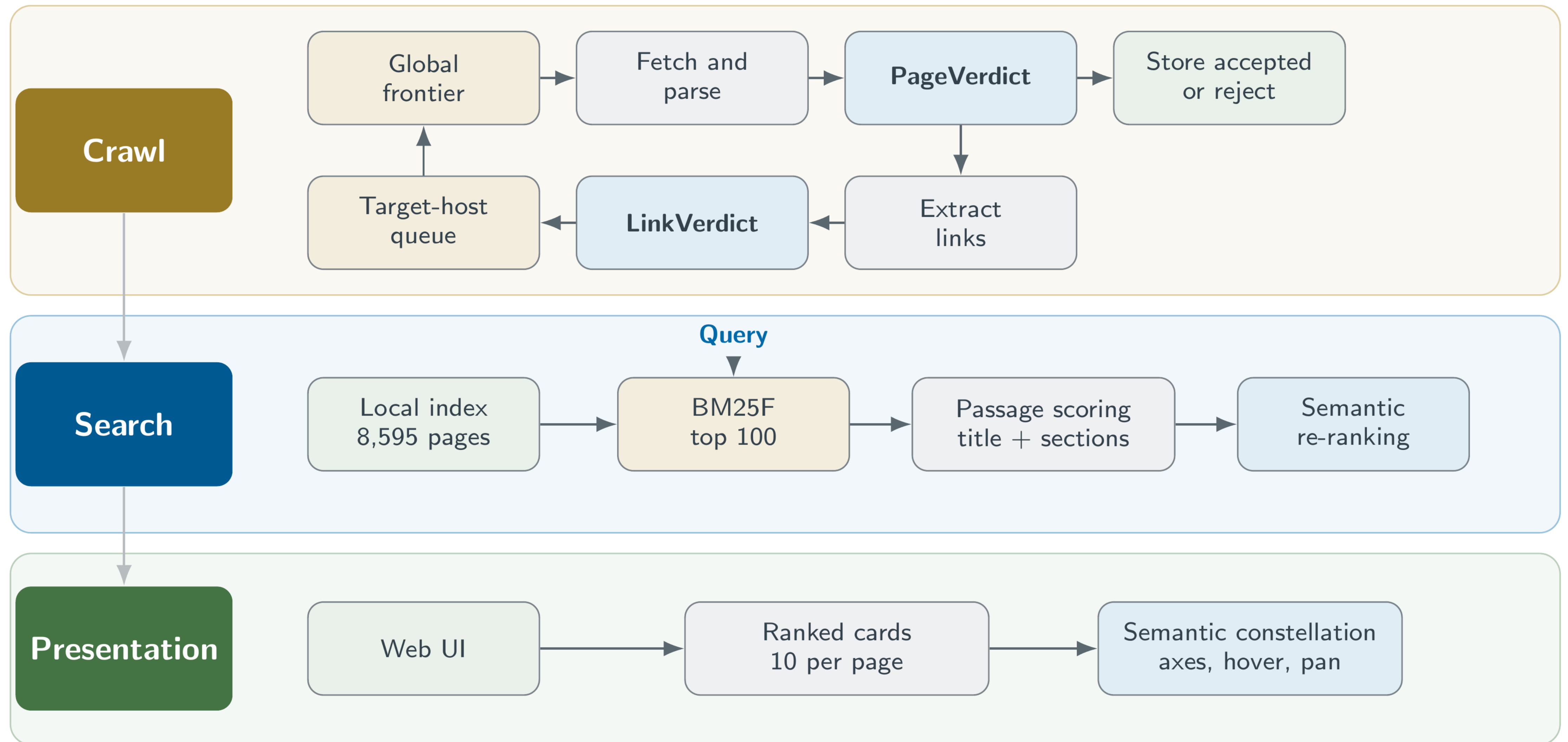


Focused Hybrid Retrieval for English Tübingen Web Content

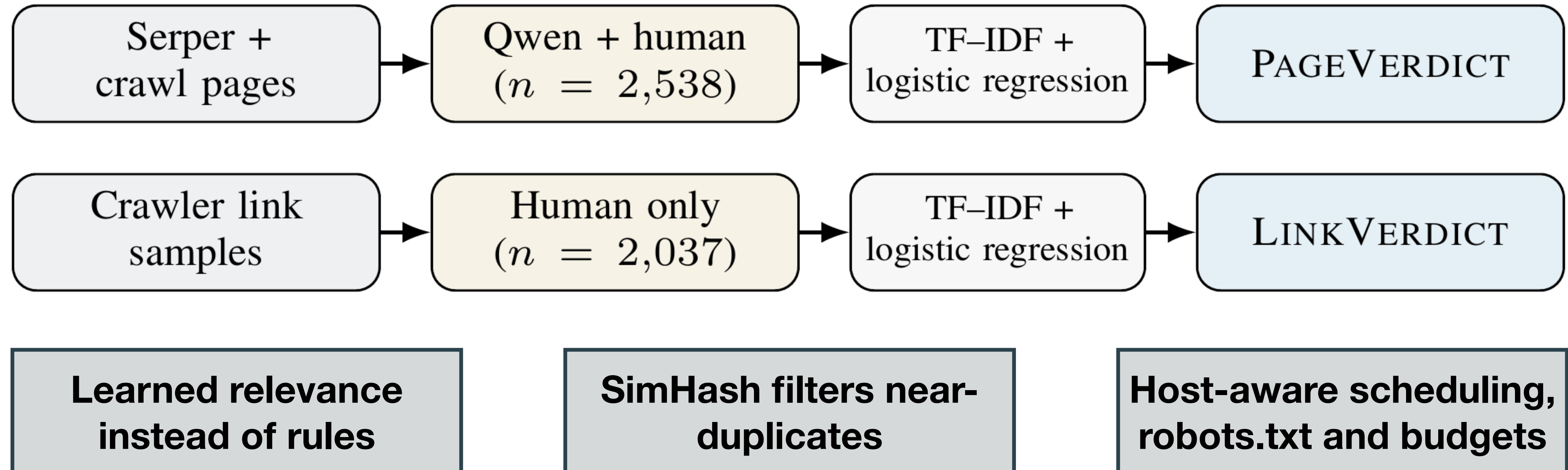
Julian Jurcevic Arne Hobrlant Andreas von Bendemann

System Overview



Teacher-Student Architecture

Crawl

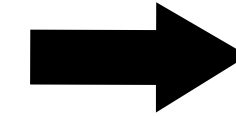


Final Crawl

- ▶ **Fetches** 23,943 pages
- ▶ **Accepted** 10,028 pages
- ▶ **Indexed** 8,595 English pages (language safeguard)

► Lexical score s_{BM25F}

- BM25F score
 - Field weights differ for body, title and URL
- Retrieves top 100 documents



► Semantic score s_{sem}

- Passage level scoring, prevents dilution of long pages
- Up to 20 overlapping passages per page
- Embed title separately
- Best and top-3 passages

► Final score s_{final}

- $s_{\text{final}}(d, q) = \alpha \hat{s}_{\text{BM25F}}(d, q) + \beta \hat{s}_{\text{sem}}(d, q)$,
with $\alpha = 0.4$ and $\beta = 0.6$

► Development Set

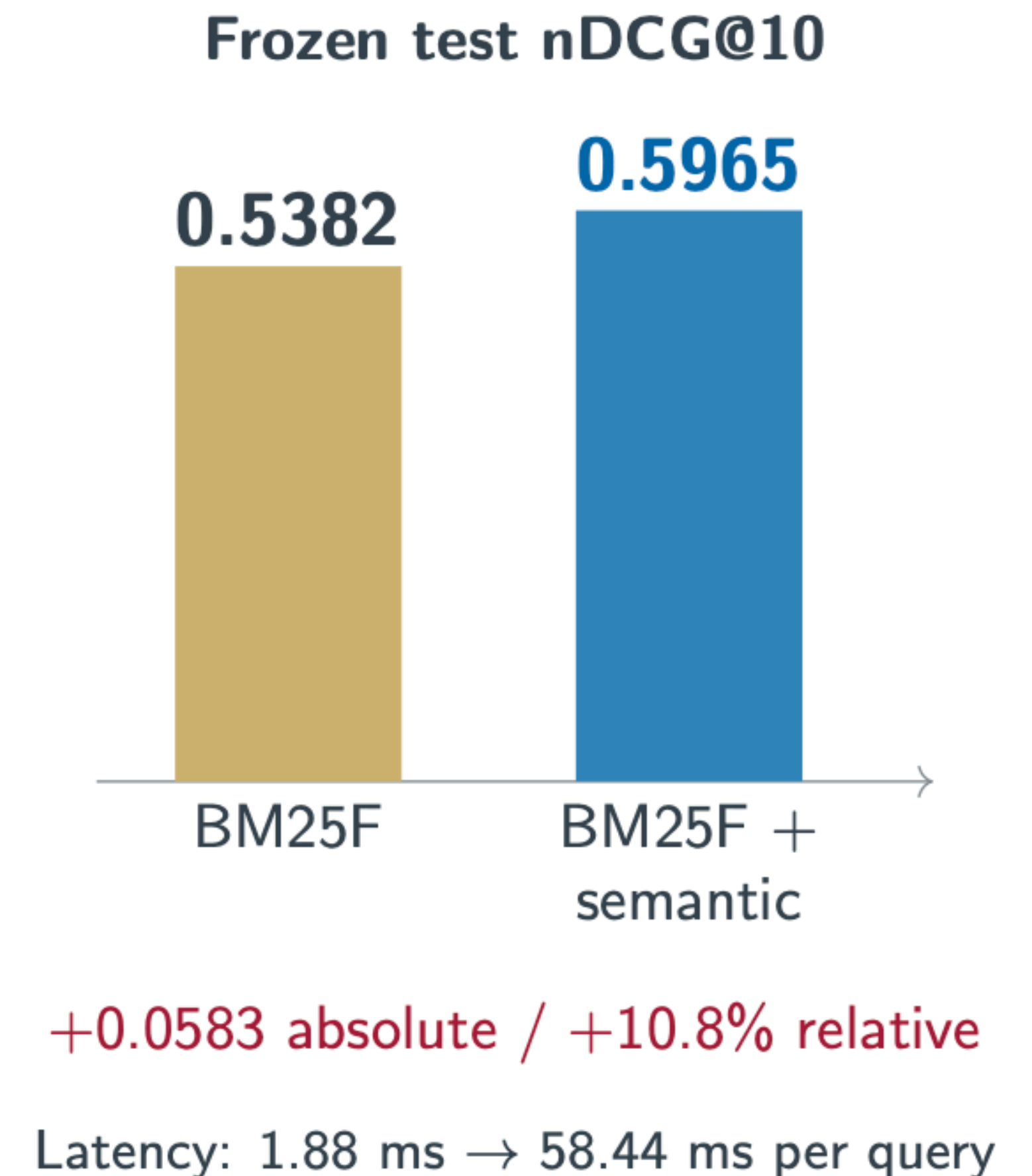
- Consists of 20 queries (various topics)
- Retrieve top-100 documents based on queries
- Judge based on query, retrieved documents from (0-3)
- Select configuration with best nDCG@10

► Test Set

- Consists of 10 queries (various topics)
- Judge Performance of selected configuration

► Result

- Semantic re-ranking improved score
- Proximity scoring did not improve score



Demo

Any Questions?

Modern Search Engines – SS 2026 Final Presentation
